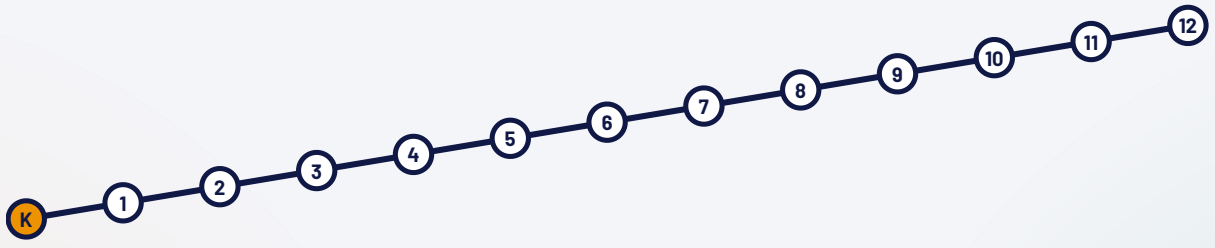
Awsaj Academy

K-12 VERTICAL ALIGNMENT · AN ANALYTICAL WALKTHROUGH

Our Curriculum Journey: Math

What our teachers' ratings tell us about how our standards are taught, K-12, and where we could go next.



From the ratings you entered as teachers (2,709 ratings, November 2026): what was reported as taught, not what students know.

Contents

1	**Math, K-12** Math from Kindergarten to Grade 12.	3
2	**About the data (at the back)** Where the data comes from, how to read the ratings, the words we use, the limits.	12

Each part: the picture (curricular capacity, grade by grade), then Facts, Patterns and Moving forward, five items each. Items link to the Vertical Alignment site. About the data is at the back. 14 pages, A4.

Taught in depth Introduced / exposed Not taught

MATH, KINDERGARTEN TO GRADE 12

Math, K-12

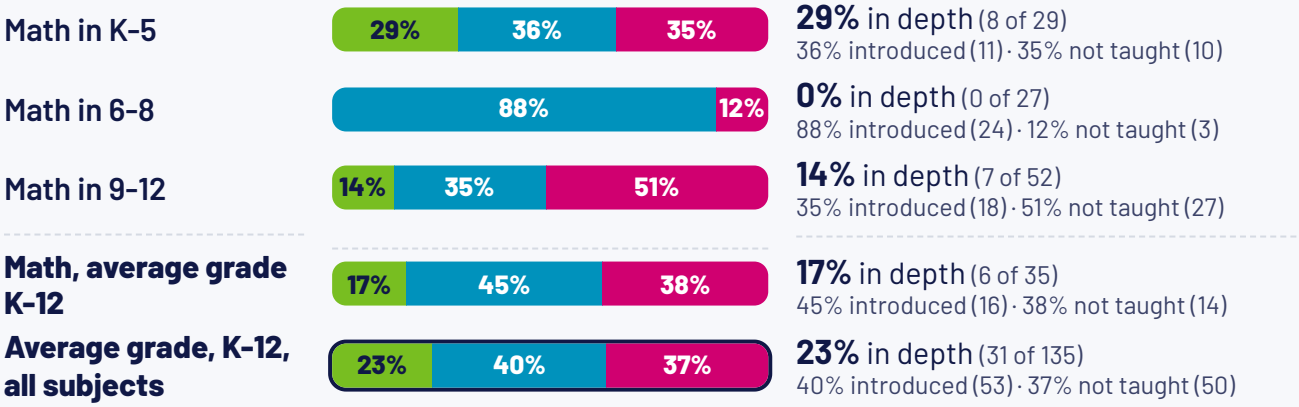
Across K-12, teachers reported that 62% of the 460 Math ratings (406 standards) are at least introduced, and 17% are taught in depth. Math has real peaks: Grade 2 (62% in depth), place value in K-5 (60%) and Grade 9 equations (60%).

THE PICTURE

At a glance: our curricular capacity

What you are seeing. An average grade in a year, by subject: green in depth, blue introduced, pink not taught, with the number of standards.

What it adds to our story. Introduced is Math's most common rating (45%). Depth is strongest in K-5 and in Grade 9.



Curricular capacity: what an average grade reports taking to depth, introducing or not teaching in a year. From 460 grade-level ratings, Essential Elements apart. Not a target.

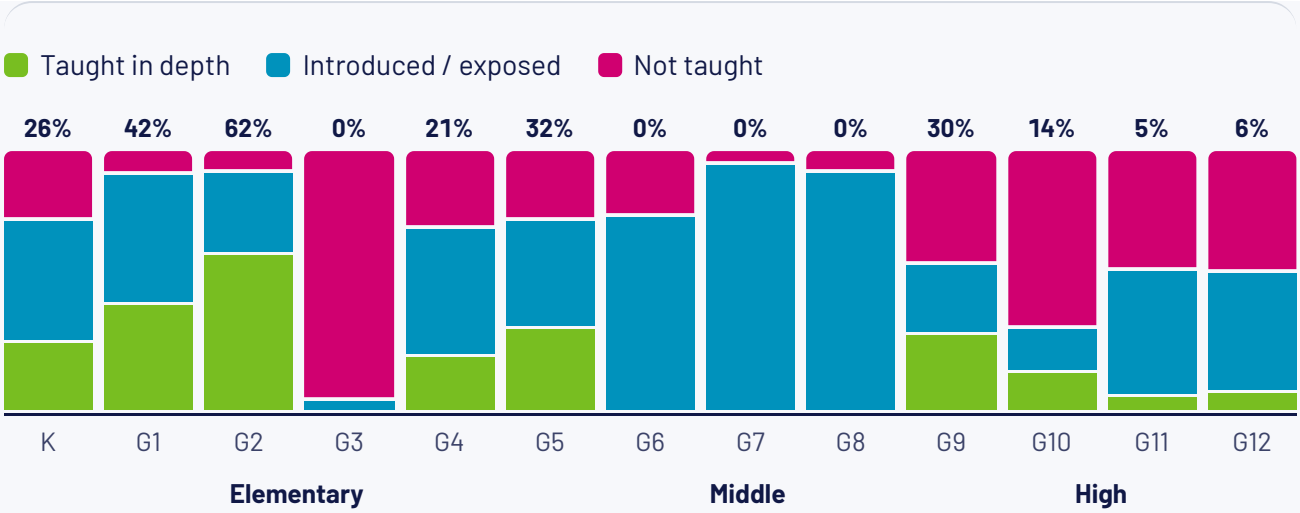
Open the coverage map on the site ↗

THE PICTURE

Grade by grade

What you are seeing. One column per grade. The top number is the share taught in depth.

What it adds to our story. Strong early grades and a strong Grade 9 frame the middle years, where Math is introduced rather than taken to depth.


Taught in depth
Introduced / exposed
Not taught
26%
42%
62%
0%
21%
32%
0%
0%
0%
30%
14%
5%
6%
K
G1
G2
G3
G4
G5
G6
G7
G8
G9
G10
G11
G12
Elementary
Middle
High

Strand	K	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12
Counting and Cardinality	25%												
Geometry	50%	33%	33%	0%	100%	25%	0%	0%	0%				
Measurement and Data	33%	50%	20%	0%	29%	38%							
Number and Operations in Base Ten	0%	63%	100%	0%	17%	75%							
Operations and Algebraic Thinking	0%	22%	100%	0%	17%	33%							
Number and Operations - Fractions				0%	0%	0%							
Ratios and Proportional Relationships							0%	0%					
Statistics and Probability							0%	0%	0%				
The Expressions and Equations							0%	0%	0%				
The Number System							0%	0%	0%				
Functions									0%				
Arithmetic with Polynomials and Rational Expressions										0%		0%	50%
Building Functions										0%		0%	0%
Creating Equations										50%	100%	0%	0%
Interpreting Categorical and Quantitative Data										11%			
Interpreting Functions										33%		33%	13%
Linear, Quadratic, and Exponential Models										0%		0%	0%
Quantities										0%	100%		
Reasoning with Equations and Inequalities										60%	100%	17%	0%
Seeing Structure in Expressions										0%		0%	50%
The Real Number System										100%			
Circles											0%		
Conditional Probability and the Rules of Probability											0%		0%
Congruence											31%		
Expressing Geometric Properties											0%		0%
Geometric Measurement and Dimension											0%		

Strand	K	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12
Similarity, Right Triangles, and Trigonometry											0%		0%
Making Inferences and Justifying Conclusions												0%	0%
The Complex Number System												0%	11%
Trigonometric Functions												0%	0%
Using Probability to Make Decisions												0%	
Vector and Matrix Quantities													0%

How to read this: each square is the share of that strand's rated standards taught in depth in that grade. Darker green means more in depth. A blank square means the grade has no standards in that strand.

Open the grade dashboards ↗ Gaps and repetition ↗ Progressions ↗



SECTION 1 OF 3

Facts

What you are seeing. The five main facts. Each links to its data on the site.

What it adds to our story. Number is our strength. Fractions, ratios and equations, the path to algebra, are the thread to follow.

1. BRIGHT SPOTS **Number work and Grade 9 equations stand out.** Place value is 100% in depth in Grade 2, 75% in Grade 5, 63% in Grade 1. Grade 2 Operations and Algebraic Thinking (4 of 4), Grade 4 Geometry (3 of 3) and Grade 9 The Real Number System are fully in depth. Coverage map ↗

2. GAPS **Fractions are not in depth in Grades 3 to 5.** 0% in depth across Grades 3 to 5. Not taught: 100% in Grade 3, 42% in Grade 4, 64% in Grade 5. Coverage map ↗

3. GAPS **70 standards are not taught in their own grade.** 60 are in K-5 and 10 in 6-8, most in Fractions (19), Operations and Algebraic Thinking (15) and Measurement and Data (15). In high school, 77 standards are not taught in the course where they fit best. Gaps and repetition ↗

4. GAPS **10 of 18 progressions break down.** Each breaks down on an official Coherence Map link (a published map of which Math standards build on which); ratios is taught at every step. Three Grade 9 standards in depth build on Grade 6 or 8 standards rated not taught (official links). Progressions ↗

5. BREADTH WITHOUT DEPTH **186 standards are introduced, never in depth.** 63 in K-5, 71 in 6-8 and 52 in 9-12. Gaps and repetition ↗



Patterns

What you are seeing. What the facts show side by side. Possibilities to discuss, not conclusions.

What it adds to our story. Following one idea, such as fractions to equations, across the grades shows where teaching may need more depth.

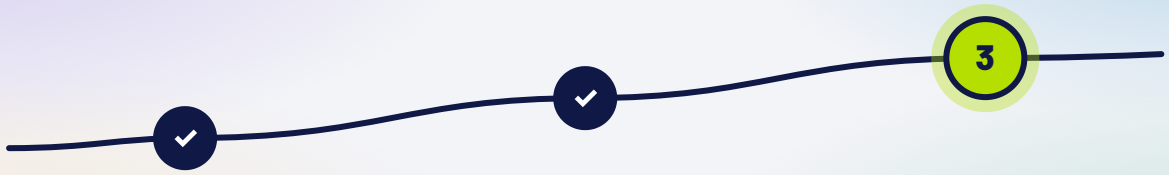
1. HANDOFFS **The road from arithmetic to algebra is thin.** Fractions (Grades 3 to 5), ratios (introduced only in Grades 6 and 7) and expressions and equations have no standard rated in depth from Grade 3 to Grade 8. Each depends on the one before. Gaps and repetition ↗

2. HANDOFFS **Grade 3 is a hinge year.** Multiplication, division, fractions and area all begin in Grade 3. Six progression findings land there, the most of any grade in Math, and 4 of the 10 broken progressions trace back to Grade 3 standards rated not taught. Progressions ↗

3. HOW WE RATED **Whole grades rated almost the same way.** Grade 3 rated 31 of 32 standards not taught, and middle school 71 of 81 introduced (its Essential Elements 7% in depth). Uniform ratings across a grade can say as much about the reading of the scale as about each standard. Grade 3 dashboard ↗

4. ESSENTIAL ELEMENTS **K-5 Essential Elements tell a second story.** 82% of K-5 Math Essential Elements are in depth (95% in Grades 1 and 5). Deep Math teaching for students working below grade level may be happening at that level, which the grade-level ratings do not show. Coverage map ↗

5. TOWARD THE SAT **High school Math narrows by design or by default.** Vectors, complex numbers, trigonometric functions and probability for decisions have little or nothing; several are advanced strands not every pathway includes. On the SAT road, algebra is strongest (about 28% in depth), data analysis (about 3%) and geometry and trigonometry (about 7%) thinner. Coverage map ↗



SECTION 3 OF 3

Moving forward

What you are seeing. Five questions for teams, each with its data and a first step we could take.

What it adds to our story. A few Math standards to take deeper together, across grades.

1

IMPROVING THE CURRICULUM **Which 10 Math standards per grade would we protect if we could only take 10 to depth?**

(Data: 17% of Math in depth K-12; 186 standards introduced and never in depth) see Fact 5

First step: We could agree a short list of Math standards per grade that every class takes to depth, protected in the timetable.

Gaps and repetition ↗

2

IMPROVING THE CURRICULUM **Which high school strands have we chosen not to teach, and does each student's pathway still reach what the SAT asks?**

(Data: 51% of high school Math not taught; 11 of 21 strands have nothing in depth; 77 standards not taught in the course where they fit) see Fact 3

First step: We could name the high school strands we have chosen not to teach and check each pathway against what the SAT asks.

Gaps and repetition ↗

3

WORKING ACROSS GRADE LEVELS **What would Grade 5 and Grade 6 teachers each want the other to see in their classrooms?**

(Data: Grade 2 62%, Grade 3 0%, Grade 5 32%, Grades 6 to 8 0%, Grade 9 30% in depth) see Fact 2

First step: We might hold one moderation session for Grades 2, 3, 5, 6 and 9, each bringing a piece of student work on the same idea.

Coverage map ↗

4

KNOWING THE BANDS ABOVE AND BELOW **If we followed one student's path through fractions, ratios and equations from Grade 3 to Grade 9, where would it feel secure and where would it feel thin?**

(Data: Fractions 0% in depth in Grades 3 to 5; ratios and expressions 0% in Grades 6 to 8; Grade 9 equations 60%; 3 official hand-over gaps into Grade 9) see Fact 2

First step: We could follow fractions, ratios and equations as one thread from Grade 3 to Grade 9, with each grade naming its part.

Progressions ↗

5

HOW WE RATE AND PLAN **What did each team mean by its ratings, and would all five teams read the scale the same way after talking it through together?**

(Data: Grade 3 rated 31 of 32 not taught; middle school 71 of 81 introduced; 45% of Math ratings are introduced) see Pattern 3

First step: We might talk the scale through together, with evidence of what "in depth" looks like in every class: a shared quiz, a common task, reteaching time.

Grade 3 dashboard ↗

WHERE THE NUMBERS COME FROM AND HOW TO READ THEM

About the data

What our teachers reported about how our standards are taught, K-12. Eight parts: the whole school, the three grade bands and the four subjects. Each part opens with the picture (curricular capacity, grade by grade), then Facts, Patterns and Moving forward, five items each.

ABOUT THE DATA

How to read this paper

> **What you are seeing.** How the ratings were made, what the three ratings mean, and the words we use.
>
> **What it adds to our story.** It helps us read every number fairly: these are teachers' reports of teaching, not measures of what students know.

What our teachers reported about how our standards are taught, K-12. Eight parts: the whole school, the three grade bands and the four subjects. Each part opens with the picture (curricular capacity, grade by grade), then Facts, Patterns and Moving forward, five items each.

Where the numbers come from. The ratings sheet of 9 November 2026 (2,166 standards, 2,709 ratings), checked against the Vertical Alignment reports.

The three ratings.

Rating	What it means
Taught in depth	ALL classes in the grade consistently teach, reteach, review and assess the standard
Introduced / exposed	Anything in between (for example, 3 of 5 classes)
Not taught	No class in the grade teaches it

Keep in mind.

- The ratings report teaching across a whole grade, not what students know, and they judge no one.
- "In depth" is all or nothing. Teams may have read it more or less strictly.
- Much literacy is also taught in Science and Social Studies, and may not show in the ELA ratings.
- Shares are of grade-level ratings. Essential Elements are counted apart.

Curricular capacity. What an average grade reports taking to depth, introducing or not teaching in a year. Not a target: we cannot teach everything equally, so we choose the mix on purpose.

Categories. Each item carries a small label (for example Bright spots, Gaps, Handoffs), the same set in every part, so a team can follow one kind of observation across parts.

ABOUT THE DATA

Method, sources and limits

> **What you are seeing.** How the ratings were made, what the three ratings mean, and the words we use.
>
> **What it adds to our story.** It helps us read every number fairly: these are teachers' reports of teaching, not measures of what students know.

Every figure in this paper was computed from the ratings sheet and cross-checked against the site. Where the site's band and K-12 views differ (progressions), the paper says which it uses. Percentages are rounded half up, as on the site.

What was read

Source	How it was used	How fully it was read
Ratings sheet, "All Standards and Ratings K-12" (9 Nov 2026)	Every figure	All 2,166 rows
Organized ratings workbook	Cross-check of totals	All tabs
20 Findings decks (band and subject)	Cross-check, questions and next-step themes	All slides and speaker notes
K-12 Reports PDF (118 pages)	Cross-check	First 80 pages scanned by search; a few read directly
7 other Reports PDFs	Cross-check of headline figures	Scanned by search, not read page by page

How terms are used

- **Rated standards.** One rating is one standard in one grade. A standard shared by several grades is counted once for each grade that rated it.
- **At least introduced.** Rated in depth or introduced.
- **Not taught where it belongs (confirmed).** The grade the standard belongs to rated it not taught (for a standard shared across grades, every grade that shares it did). In Math and ELA each standard belongs to one grade, so this means that grade's team rated it not taught; it does not mean other grades should have taught it.

- **Not taught where it fits.** Each standard was placed in the grade or course where it fits best, and that grade or course rated it not taught. Other grades were not asked to rate it.
- **Introduced, never in depth (met but not owned).** Introduced in at least one grade and in depth in none.
- **Progression finding.** An automatic check on a chain of linked standards: a step not taught, or a standard in depth while the one it builds on is not taught.
- **Taught at every step.** Every standard in the chain, across two or more grades, is at least introduced.
- **Progression.** A chain of linked standards across grades. Official links are the Coherence Map (Math), anchor standards (ELA) and the standards' own structure (Science, Social Studies). Other links are suggestions.
- **Curricular capacity.** The average number of standards one grade rated in depth, introduced and not taught in a year (all ratings divided by the number of grades), rounded.
- **Hand-over gap.** A standard taught in depth while a standard it builds on, placed in an earlier school, is rated not taught.
- **Priority and power standards.** Standards the school's own documents name: power standards in K-5, priority standards in 6-12.

Limits to keep in view

- Ratings are teachers' judgements of teaching across a grade. They are not observations and not student results.
- Grades may have read "in depth" differently. Uniform ratings across a whole grade are a reason to check how the scale was read; the paper points these out so teams can discuss them.
- Each team rated its own subject. Literacy taught in Science and Social Studies, and Math taught in Science, may not show in the ELA and Math ratings.
- Middle school Science and high school Math, Science and Social Studies standards were placed in the grade or course where each fits best. 24 K-5 Social Studies standards are also rated not taught in the grade where they fit.
- Band counts of progressions look only at steps inside the band, so they do not add up to the K-12 figure.
- Literacy audit figures come from the 2026 PK-5 literacy audit of weekly plans.
- Some strands have few standards in a grade, so a single rating can move a percentage a long way. Counts are given where this matters.
- Suggested links between standards (most ELA, Science and Social Studies hand-overs) have not been checked by teachers.

Status. This is a draft for the Vertical Alignment Committee. Each finding should be checked with the teams concerned before it is treated as settled. The planned review of weekly and unit plans (Part 2) will test many of these patterns against what is actually planned.